

ACC Two-Channel Gate-Structured Architecture

Physical Conditions

$|\Delta x|, \text{EMA}(v), \theta, \dot{\theta}$

F_z^L, F_z^R

$h_{\text{ground}}^L, h_{\text{ground}}^R$

Wheel Torque τ_w

Leg Torque τ_q

gate inputs

τ_{balance}

$\tau_{\text{posture}}(h^{\text{cmd}})$

g_{anchor}

τ_{anchor}

g_{terrain}

$\Delta\tau_{\text{post}}$

g_{flight}

τ_{flight}

$$\tau_w = \tau_{\text{bal}} + g_a \tau_a + g_f \tau_f$$

$$\tau_q = \tau_{\text{post}} + g_t \Delta\tau$$